News Media Signals for Predicting Valley Fever Case Rates

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Abstract. Endemic to California's Central Valley and southwestern parts of the United States, Valley Fever is a respiratory disease that may be influenced by various environmental factors. Since the disease is prevalent in areas under arid climate conditions, prior research focuses on how environmental conditions predict future case reports. However, there is limited research on how public awareness and news media coverage plays a role in these predictions. This work examines how Google News articles mentioning terms related to Valley Fever are associated with monthly case rates in Fresno and Kern counties, two high-incidence counties in the Central Valley. To evaluate predictive accuracy, a baseline regression, seasonal naive, Long Short-Term Memory, and SARIMAX models are compared. We find that news articles on Valley Fever containing risk-related terminology and regional mentions of Central Valley counties are correlate with reported case patterns across both counties, even when accounting for traditional environmental predictors.

Keywords: Valley Fever · News Media Analysis · Natural Language Processing · Long Short-Term Memory · Time-Series Forecasting

1 Introduction

1.1 Valley Fever as a Public Health Concern

Coccidioidomycosis, also known as Valley Fever, is a fungal disease that has become a growing public health concern for the Southwestern parts of the U.S. It is most prevalent in semi-arid regions where there is a severe lack of available water, dusty winds, and dry conditions [17, 7]. The disease accompanies these conditions because fungal spores can become airborne when dry soil is disturbed, which then increases the risk of inhalation and infection [15]. Although it cannot be contracted interpersonally, it is still a public health concern due to increasing Valley Fever incidence rates and the spread of the disease beyond historically endemic areas [12].

Recent research suggests that climate change may further contribute to the spread of Valley Fever. In a recent study, researchers found that increased temperatures and prolonged dry conditions due to climate change may increase the frequency of dust storms and soil disturbance events [12]. These events allow fungal spores to become airborne through the wind and transported across larger geographic regions outside of just the Central Valley. This exemplifies how environmental and regional conditions may influence Valley Fever exposure risk, which motivates the use of environmental predictors within forecasting models for future case rates.

Prior research has incorporated environmental factors such as precipitation, fire, and air quality into time-series forecasting models to predict Valley Fever case rates [20, 5]. More recently, LSTM models have improved prediction of future case rates [20, 11, 9]. These environmental and regional conditions motivate the use of environmental predictors in forecasting models, but they also raise the question of whether non-environmental factors, such as public awareness and news media coverage, contribute to reported case rates.

1.2 Public Awareness and Media Coverage

Beyond environmental conditions, it is also important to consider other external factors that may contribute to case reporting of the disease such as misdiagnosis. Since symptoms of the disease frequently resemble other respiratory illnesses such as influenza or COVID-19, it is often not reported initially as Valley Fever [6]. This may result in delays in diagnosis, which can skew case reporting timelines that lag behind the actual onset of symptoms. The delays and misdiagnosis may coincide with low patient awareness, as individuals who are not familiar with the disease are unlikely to prompt their physicians to test for it.

Previous survey research shows that the lack of patient awareness is widespread, with less than half of Californians during 2016 and 2017 stating that they had heard of Valley Fever. Moreover, only 25% of respondents living in high-incidence regions, such as Fresno and Kern County, were aware that they lived in areas where fungal spores exist [10]. This demonstrates a large awareness gap and motivates the question of whether public awareness may play a role in reporting.

Since there is limited research on whether awareness through news media coverage on Valley Fever contributes to case reporting, we explore the relationship in this study. Specifically, we scrape Google News articles mentioning terms related to Valley Fever and extract monthly media features such as news article count, regional mentions, and risk-related terminology. Looking at the content within each news article allows us to evaluate how the severity of the disease is communicated to the public and whether media urgency corresponds to changes in reported case rates across Fresno and Kern County. To motivate this question, Figure 1 compares Kern County case rates with monthly Google News article volume. The visual pattern suggests that periods of elevated case rates are followed by increased coverage beginning in 2013, which supports using time-lagged media predictors in forecasting.

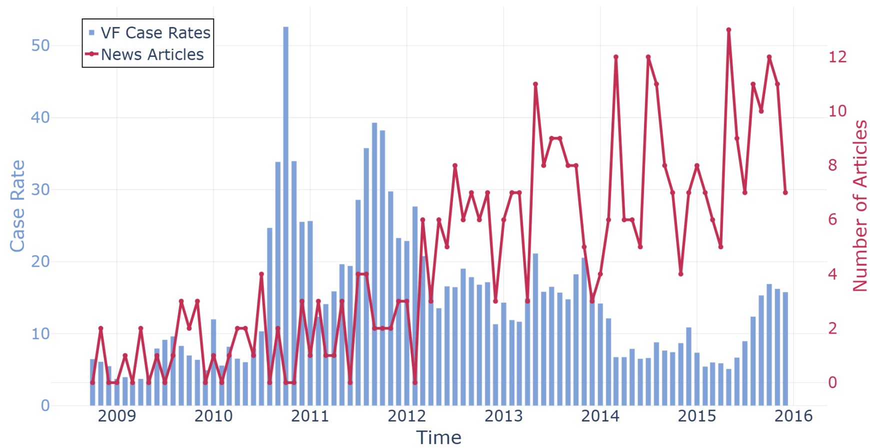


Fig. 1: Dual-axis chart of monthly Kern County Valley Fever case rates and news article volume from 2008 to 2016.

2 Methods

2.1 Valley Fever Case Rate Data

Monthly Valley Fever case counts in Central Valley counties were converted to case rates based on the number of cases per 100,000 people. The case rates were calculated using population estimates provided by U.S. Census Intercensal data and were preprocessed in a ValleyCast GitHub repository [18, 19, 13, 3]. Within this work, multiple environmental datasets were aggregated into county-level monthly observations. Environmental features include precipitation, wind metrics, Air Quality Index (AQI) Particulate Matter (PM) 2.5 and 10, earthquake totals, fungicide application totals, and wildfire acres burned. Due to Valley Fever's high-incidence regions being Fresno and Kern County, this study will center data that ranges from dates starting at the beginning of 2008 to the end of 2015 within these counties.

2.2 Google News Feature Engineering

To explore Google News media coverage on Valley Fever, the Python module `pygooglenews` [4] was used to query the number of monthly news articles mentioning terms such as "Valley Fever", "Coccidioidomycosis", and "San Joaquin Valley Fever." Since media sources frequently refer to the disease using geographic naming associated with California's Central Valley, the last search term was included to improve article retrieval.

Beyond the monthly article count, the full text from each article was extracted using the `newspaper4k` [14] Python module. Natural Language Processing techniques were used to normalize article text by removing URLs, bracket content, parentheses with citations, external source names, and white space. After this preprocessing pipeline, keyword extraction was used to look at regional mentions and risk-related language associated with Valley Fever reporting. Specifically, the number of times a Central Valley county such as Fresno, Kern, Tulare, Kings, Madera, Merced, and articles containing "Central Valley" were scraped. Moreover, the number of times a risk word such as "outbreak", "surge", "spike", "increase", "danger", "severe", "deadly", "hospital", or "death" were also scraped.

Risk word and regional mention frequencies were converted to rates by dividing the mention frequency over the number of articles. These media-related variables were aggregated by year and month to align with the monthly Valley Fever case rate and environmental data. To investigate whether Valley Fever case rates respond to news coverage from months prior, article volume, regional mention rates, and risk-related mention rates were time-lagged by different months. Since three months had the strongest overall model performance, a time-lag of three months was chosen.

2.3 Regression and Influence Diagnostics

Multiple linear regression models were used to evaluate the relationship of Valley Fever case rates at month t and lagged news media variables at month $t-3$. The response variable was defined as the log-transformed monthly case rate and predictor variables as the time-lagged media variables, seasonal effects, and county interaction terms. Log-transforming the response variable helped achieve model assumptions of normality and constant variance. Since the monthly news media data is equivalent across both Fresno and Kern County, the standard errors in the model were clustered by month. This allows for observations within the same month to share correlated error structure and display more reliable regression results.

Cook's Distance (D_i) and Difference of Fits ($DFFITS_i$) influence diagnostics were used to identify influential observations that disproportionately affect the regression model. Separate models were refit after removing influential observations exceeding the DFFITS threshold,

$$|DFFITS_i| > 2\sqrt{\frac{k+2}{n-k-2}}$$

and Cook's Distance threshold,

$$D_i > \frac{4}{n},$$

where k denotes the number of predictor terms and n denotes the total number of observations [16]. Comparing models helped determine whether estimated effects of news media variables remained consistent across different influence diagnostic methods.

2.4 LSTM Time-Series Forecasting

Long Short-Term Memory (LSTM) models are neural networks used to learn patterns in either the short or long term of past sequential data to predict the future [8]. In contrast to traditional neural networks that treat observations independently, LSTM models retain information from previous time steps with the use of memory cells and gating mechanisms. These gates decide which information is remembered, updated, or forgotten over time. It allows the model to capture both short-term and long-term temporal dependencies within sequential observations such as time-series data. Since Valley Fever case rates in Fresno and Kern County have seasonal outbreak patterns, LSTM models were chosen to capture temporal relationships between historical case rates, environmental variables, and news media predictors.

The inputs for this study include environmental variables from ValleyCast [3] and scraped news media predictors, with log-transformed Valley Fever case rates being the target variable. Three LSTM variants were trained for Fresno and Kern County: a combined news-plus-environment model, an environmental-only model, and a news-only model. Hyperparameters of sequence length, hidden size, dropout, learning rate, and number layers were fine-tuned from a grid search. An 85-/15% time-chronological train/test split of the data was used to evaluate forecasting performance on unseen future data.

The Root Mean Squared Error (RMSE) and the Mean Absolute Error (MAE) for each split was also analyzed to evaluate predictive performance. Moreover, time-series visualizations were used to compare the predicted and observed Valley Fever case rates across both splits. Permutation Feature Importance (PFI) helped evaluate how much each individual predictor variable contributed to LSTM forecasting performance. PFI scores are computed by randomly shuffling the values of each feature's column and calculating the increase in RMSE after the shuffle [2]. Features with a higher RMSE increase indicate that the model relied more on the feature. Since the PFI scores vary across multiple LSTM runs, the average PFI scores across 100 runs were calculated for both counties to reduce variability caused by model training.

2.5 SARIMAX Time-Series Forecasting

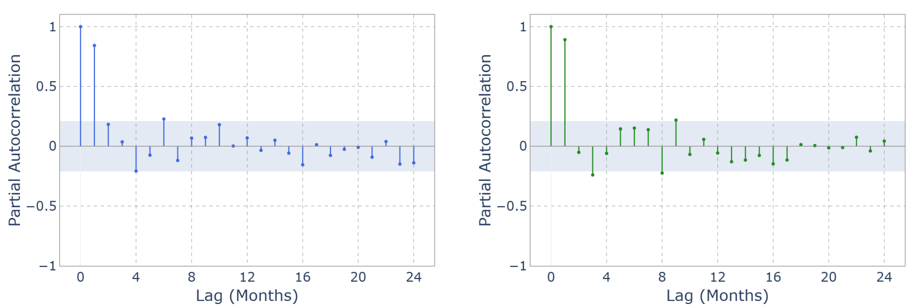
Seasonal Auto Regressive Integrated Moving Average with Exogenous Variables (SARIMAX) models build off traditional ARIMA forecasting models by specifying a seasonal structure and considering predictor variables that exogenously influence the response variable [1]. The autoregressive (AR) component incorporates the temporal dependence between current Valley Fever case rates and previous monthly observations. Moreover, the moving average (MA) component captures relationships between current values and previous forecast errors. Lastly, the integrated component ensures that the data is stationary by removing long-term trends and differencing data observations. These components correspond to parameters in the SARIMAX model, which is defined as SARIMAX$(p, d, q)(P, D, Q, s)$. The difference between the non-seasonal and sea-

Table 1: SARIMAX model parameter definitions.

Parameter	Definition
p	Non-seasonal autoregressive order
d	Non-seasonal differencing order
q	Non-seasonal moving average order
P	Seasonal autoregressive order
D	Seasonal differencing order
Q	Seasonal moving average order
s	Seasonal period length

sonal parameters are that the seasonal components model patterns that occur across patterns that repeat over a fixed time intervals. Since Valley Fever case rates have strong yearly trends that are exist alongside seasonal weather patterns, incorporating this structure in the model is important. In this study, a seasonal period length of $s = 12$ was used to represent yearly seasonality in monthly observations. Thus, the seasonal captures relationships across observations such as t, $t-12$, $t-24$, and other yearly lag intervals.

To determine other hyperparamters in the SARIMAX model, Partial Autocorrelation Function (PACF), Autocorrelation Function (ACF), and Augmented Dickey-Fuller (ADF) analyses were conducted. PACF plots were used to identify the autoregressive order p by analyzing the direct correlation between current observations and previous lagged values [1]. Similarly, ACF plots were used to determine the moving average order q by evaluating overall serial correlation across lagged observations. Lastly, the ADF test was used to evaluate whether the data was stationary and determine the differencing order d.



(a) Fresno County **(b)** Kern County

Fig. 2: Partial Autocorrelation Function (PACF) plots for monthly log-transformed Valley Fever case rates in (a) Fresno County and (b) Kern County.

The largest spikes in both county PACF plots occur at a time-lag of 1 and 2 months, suggesting a strong autoregressive dependence across recent observations of case rates. These spikes exist outside of the highlighted confidence interval, which also indicates that statistically significant partial autocorrelation occurs at lower lag values. Since the strongest direct relationship occurred at a time lag 1 month, an autoregressive order of $p = 1$ was chosen for the SARIMAX models.

In a similar ACF analysis, there were high early autocorrelation values and a decrease over larger lags. Since Fresno and Kern County showed these results, there was temporal correlation in monthly Valley Fever case rates. The captured relationship between current case rate values and previous forecast errors, from the ACF analysis resulted in the choice of a moving average order, $q = 1$ across both counties. Lastly, ADF testing showed that the log-transformed case rate data was not stationary and differencing was necessary for both counties, $d = 1$.

After conducting PACF, ACF, and ADF analyses, the final SARIMAX model was defined as SARIMAX$(1, 1, 1)(1, 0, 1, 12)$. The seasonal differencing order was specified as $D = 0$ due to the yearly seasonal structure remaining relatively stable after non-seasonal differencing. SARIMAX models were used in this study to compare whether classical statistical forecasting or deep learning methods, such as LSTM modeling, capture trends in Valley Fever case rates better. Similar to the LSTM models, the forecasting performance of the SARIMAX models in this study were evaluated using RMSE and MAE values and time-series visualizations across training and test splits. In addition to these forecasting models, a seasonal naive baseline was used for comparison. For each forecasted month, the seasonal naive model predicts the Valley Fever case rate using the observed value from the same month in the previous year, providing a benchmark for yearly seasonal persistence.

3 Results

The following results evaluate the relationship between environmental variables, news media features, and Valley Fever case rates across Fresno and Kern County.

3.1 News Media Trends

Valley Fever case rates in Kern County spiked around 2011 and 2012 before gradually declining in the following years. During this same time period, the number of Google News articles mentioning terms related to Valley Fever had a gradual increase. Overall, periods of elevated case rates were followed by higher levels of news media coverage, suggesting a delayed relationship. Although article frequency spikes do not completely align with each case rate spike, the delayed relationship motivated the choice of time-lagged news media variables within the regression and forecasting models.

3.2 Regression Results

To quantify the association between lagged news media variables and Valley Fever case rates, Table 2 reports the regression coefficients after removing influential observations using Cook's Distance and DFFITS.

Table 2: Regression coefficient estimates after removing influential observations using Cook's Distance and Difference of Fits (DFFITS) thresholds. Response variable was $\log(1 + \mathrm{VFRate})_t$, using case rates from Fresno and Kern County, with standard errors clustered by month. Significance level:* $p < 0.05$.

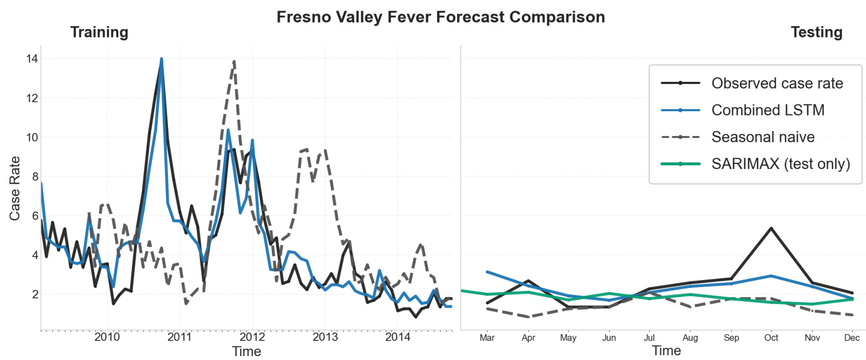
Variable	Coefficient (Cook's Distance)	Coefficient (DFFITS)
Intercept	2.0831*	2.0771*
(Number of Articles)$_{t-3}$	-0.0767*	-0.0772*
(Risk Mentions Rate)$_{t-3}$	-0.0474*	-0.0447*
(Cen Valley Mentions Rate)$_{t-3}$	-0.0341	-0.0382*
(News Articles)$_{t-3}$ $\times$ Kern	0.0834*	0.0870*
Kern County	0.7673*	0.7345*
Spring Season	-0.3800*	-0.3885*
Summer Season	-0.1927	-0.1757
Winter Season	-0.1700	-0.2087
Observations	168	173
Adjusted R^2	0.644	0.623

Across both regression models, the significance levels and coefficient estimates remained relatively stable using both influence diagnostics, which suggests that these findings were robust to those influential observations that were removed. The negative coefficient estimates for the number of news articles on Valley Fever indicate that a higher article volume from three months prior was associated with lower log-transformed case rates of the current month. Similarly, the negative coefficients for the risk-related mentions rate and Central Valley mentions rate also indicate a negative association with case rates. However, the positive interaction coefficients between the number of articles and Kern County suggest that the relationship between article volume and case rates differed in Kern County relative to Fresno County.

Beyond the news media variable coefficient estimates, Kern County in general had higher log-transformed case rates in comparison to Fresno County. Moreover, the seasonal effects show that only spring was statistically significant. The negative coefficient suggests that case rates during this season were lower in comparison to the baseline season, fall. Overall, the adjusted R^2 values across both models indicate that more than 62% of the variation in the log-transformed case rates is explained by time-lagged news media variables, county interaction terms, and seasonal effects. These results are consistent with the idea that news coverage may reflect, and possibly help signal, changes in case reporting rather than simply coinciding with them.

3.3 Forecasting Results

The forecasting comparison evaluates the combined, environmental-only, and news-only LSTM variants against SARIMAX and a seasonal naive baseline across Fresno and Kern County. The seasonal naive baseline uses the observed value from the same month in the previous year as the forecast for each month, providing a simple benchmark for yearly persistence.



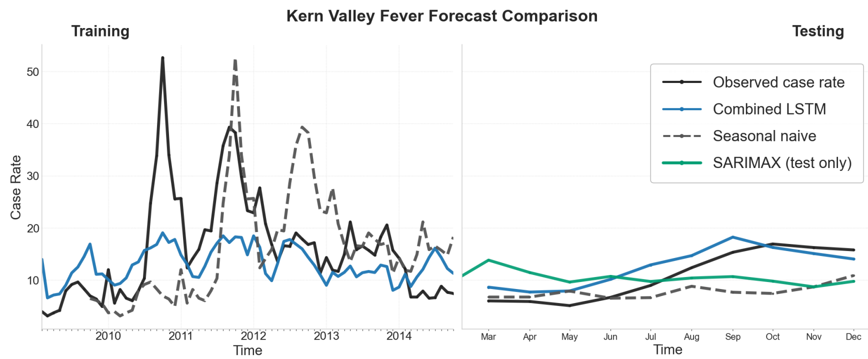
(a) Training Split **(b)** Testing Split

Fig. 3: Fresno County comparison of the LSTM, SARIMAX, and seasonal naive benchmark models for Valley Fever case rate prediction.

The Fresno County LSTM forecasts generally tracked the temporal trend in the training split (see Fig 3). In the testing split, the combined LSTM predictions were more consistent than the SARIMAX predictions, but they still captured the overall magnitude of the held-out observations despite the limited number of test points, which suggests that the LSTM model generalized to unseen data.

For Kern County, the training predictions did not track the observed values more closely than they did for Fresno County, as seen in Fig. 4. The testing split demonstrated the effectiveness of the combined LSTM, with the increase in case rates around April 2015 captured reasonably well. Moreover, the LSTM combined model reproduced the general magnitude of the test observations.

Across both counties, the combined LSTM generally produced the strongest test performance, while the environmental-only model remained competitive in Fresno County. The seasonal naive baseline was consistently weaker than the other models, suggesting that the LSTM variants captured structure beyond simple seasonal persistence. Tables 3 and 4 summarize the best model for each family. Overall, the LSTM variants tend to better capture nonlinear temporal patterns, but SARIMAX remains a competitive seasonal baseline. Hyperparameters were selected using the medium grid-search preset, which explored sequence length {2, 3, 4, 6}, hidden size {16, 32, 64}, dropout {0.0, 0.2, 0.4}, learning rate {0.0005, 0.001, 0.005}, batch size {4, 8}, and LSTM layers {1, 2}.



(a) Training Split (b) Testing Split

Fig. 4: Kern County comparison of the LSTM, SARIMAX, and seasonal naive benchmark models for Valley Fever case rate prediction.

Table 3: Fresno County forecasting metrics using the best (after hyperparameter search) model for each family.

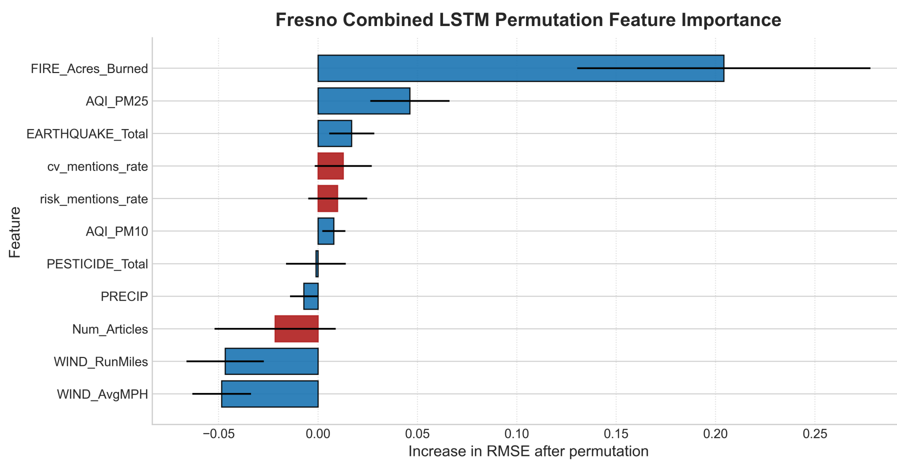
Model	Train		Test	
	RMSE	MAE	RMSE	MAE
Combined LSTM	**1.2103**	0.9568	**0.9705**	0.6283
Environmental LSTM	2.9148	**0.9424**	2.0400	**0.6170**
News-only LSTM	2.1015	1.1387	1.4860	0.7801
SARIMAX	1.6020	1.2512	1.1690	0.9049
Seasonal naive	3.5659	1.3758	2.7518	1.0402

Table 4: Kern County forecasting metrics using the best (after hyperparameter search) model for each family.

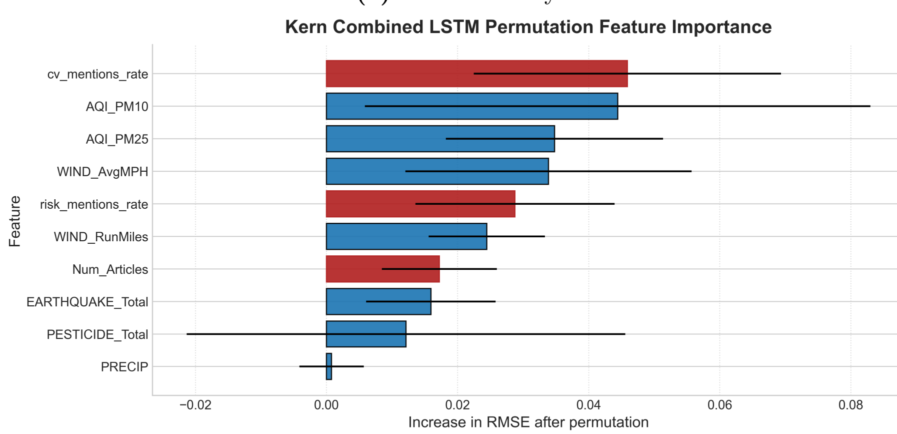
Model	Train		Test	
	RMSE	MAE	RMSE	MAE
Combined LSTM	8.5294	**2.5321**	6.3532	**2.3403**
Environmental LSTM	9.7917	2.5535	6.8930	2.3456
News-only LSTM	9.4459	3.4168	6.9969	2.8475
SARIMAX	**5.3260**	4.8686	**3.4524**	4.2498
Seasonal naive	12.0218	6.0467	8.7552	5.0161

3.4 Permutation Feature Importance

The PFI results for the combined LSTM variant show that environmental variables remained the strongest predictors in both counties, while the news-derived variables, especially the Central Valley mention rate in Fresno County and the risk word mention rate in Kern County, also contributed meaningful predictive signal. The number of Google News articles on Valley Fever had a negative PFI score in both counties, which may reflect correlated predictors or instability introduced by feature permutation. In Fig. 5, the bars indicate the mean increase in RMSE across 100 permutations, and the error bars show one standard deviation across those repeats.



(a) Fresno County



(b) Kern County

Fig. 5: Permutation feature importance (PFI) for the best-performing combined LSTM model in (a) Fresno County and (b) Kern County. Bars show the mean increase in RMSE over 100 random permutations of each feature on the held-out test set, and error bars show one standard deviation across those repeats. News-derived variables are highlighted in red.

4 Discussion and Conclusion

4.1 Media Coverage and Model Implications

This project examines whether Valley Fever news coverage contributes to case reporting by extracting Google News article frequencies, Central Valley county mentions, and risk-related terminology using natural language processing techniques. Due to the high incidence rates in Fresno and Kern County, their case rates were used as the target variable in baseline regression, LSTM, and SARIMAX models. Adding news media features beyond the environmental variables used in prior research suggests that coverage may influence case rates across both counties.

The three-month-lagged media variables from the regression models were negatively associated with log-transformed case rates, which may indicate that coverage functions as a public health warning. When news coverage increases, people in Fresno and Kern County may take preventive measures such as wearing masks during dust storms, with risk-related terminology emphasizing the severity of the disease. However, this pattern could also reflect seasonal overlap, since media coverage may peak during dry, dusty months and decline in the subsequent winter months. The seasonal effects from the regression models support this seasonal structure in Valley Fever case rates, with spring having significantly lower case rates relative to fall.

4.2 Seasonality and Forecasting Implications

Seasonality of Valley Fever outbreak patterns was captured in the LSTM and SARIMAX forecasting models using media-related variables from the regression models and environmental conditions from prior research. The SARIMAX framework provided more of a seasonal forecasting structure with less variability in the testing split, while LSTM models captured more nonlinear outbreak variability overall. The LSTM model outperformed SARIMAX in the testing split, and this suggests that LSTM modeling provides a modest improvement in predictive performance while SARIMAX remains a competitive classical forecasting method.

The PFI scores showed that environmental conditions remain critical to predicting monthly Valley Fever case rates, with the media-related variables such as Central Valley mention rates and risk-related terminology also contributing high predictive power to the LSTM models. The strong rankings of these variables suggest that the severity and media urgency communicated within Google News articles may correspond to changes in Valley Fever case rates across Fresno and Kern County. However, it's important to acknowledge that limiting the models to only two counties within California's Central Valley may reduce generalizability to other endemic regions. Moreover, Google News articles may not fully represent broader public awareness, since individuals may receive information from other media platforms, healthcare providers, or word-of-mouth.

4.3 Limitations and Future Work

This work is limited to Google News coverage, county-month aggregation, and to a small set of engineered media features, so it does not capture broader public awareness or more nuanced language patterns that may appear in other news sources or social media platforms. We also note that the results should be interpreted as associative and predictive rather than causal, since the observed relationships may also reflect seasonality, reporting delays, or correlated environmental conditions. Including other endemic regions of the Southwestern parts of the U.S. would also improve generalization for these models.

In the future, we intend to extend the media component beyond keyword counts and simple mention rates by using more advanced natural language processing methods. Sentiment or stance classification could also help distinguish neutral reporting from language that conveys uncertainty or increased risk. Ultimately, this study demonstrates that combining environmental forecasting and news media analysis can provide additional context for understanding Valley Fever reporting patterns in California's Central Valley.

References

1. Alharbi, F.R., Csala, D.: A seasonal autoregressive integrated moving average with exogenous factors (SARIMAX) forecasting model-based time series approach. Inventions **7**(4), 94 (2022). https://doi.org/10.3390/inventions7040094
2. Altmann, A., Tolosi, L., Sander, O., Lengauer, T.: Permutation importance: A corrected feature importance measure. Bioinformatics **26**(10), 1340–1347 (2010). https://doi.org/10.1093/bioinformatics/btq134
3. Báñuelos, M.: Valleycast: Environmental data and processing aggregate cases (2026), https://github.com/MBanuelos/ValleyCast, gitHub repository
4. Bugara, A.: pygooglenews: A python package for google news rss feeds (2026), https://github.com/kotartemiy/pygooglenews, gitHub repository
5. Caraccilo, J., Garner, S., Báñuelos, M.: Estimating valley fever endemicity with environmental parameters. In: 2024 58th Asilomar Conference on Signals, Systems, and Computers. pp. 1286–1290. IEEE (2024)
6. Centers for Disease Control and Prevention: Valley fever (coccidioidomycosis) (2023), https://www.cdc.gov/valley-fever/about/index.html, accessed 2026-06-21
7. Cooksey, G.L.S., Nguyen, A., Vugia, D., Jain, S.: Regional analysis of coccidioidomycosis incidence—california, 2000–2018. Morbidity and Mortality Weekly Report **69**(48), 1817 (2020)
8. Hochreiter, S., Schmidhuber, J.: Long short-term memory. Neural Computation **9**(8), 1735–1780 (1997). https://doi.org/10.1162/neco.1997.9.8.1735
9. Huender, L., Everett, M., Shovic, J.: Valley-forecast: Forecasting coccidioidomycosis incidence via enhanced lstm models trained on comprehensive meteorological data. Journal of Biomedical Informatics **162**, 104774 (2025)
10. Hurd-Kundeti, G., Sondermeyer Cooksey, G.L., Jain, S., Vugia, D.J.: Valley fever (coccidioidomycosis) awareness — california, 2016–2017. Morbidity and Mortality Weekly Report **69**(42), 1512–1516 (2020). https://doi.org/10.15585/mmwr.mm6942a2

11. Jin, X., Wei, F., Kandala, S.S., Umesh, T., Steele, K., Galgiani, J.N., Laubichler, M.D.: Time series forecasting of valley fever infection in maricopa county, az using lstm. The Lancet Regional Health – Americas **43** (2025)
12. Kumar, P., Jayan, J., Jena, D., Rai, N., Sah, S., Mawejje, E.: Valley fever on the rise: How climate change is fueling a public health crisis. New Microbes and New Infections **62**, 101500 (2024), https://pmc.ncbi.nlm.nih.gov/articles/PMC11491959/
13. Li, J., Báñuelos, M., Uminsky, D.: Valleycast: Multi-source environmental lstm modeling for early warning of valley fever. In: Accepted to the 34th European Signal Processing Conference (EUSIPCO 2026) (In press)
14. Paraschiv, A.: newspaper4k: Article scraping and curation for python (2026), https://github.com/AndyTheFactory/newspaper4k, gitHub repository
15. Pearson, D., Ebisu, K., Wu, X., Basu, R.: A review of coccidioidomycosis in california: exploring the intersection of land use, population movement, and climate change. Epidemiologic Reviews **41**(1), 145–157 (2019)
16. Pennsylvania State University: Stat 462: Applied regression analysis – identifying influential data points (2026), https://online.stat.psu.edu/stat462/node/173/
17. Stewart, E.R., Thompson, G.R.: Update on the epidemiology of coccidioidomycosis. Current Fungal Infection Reports **10**, 141–146 (2016)
18. United States Census Bureau: County intercensal population totals: 2010-2020. https://www.census.gov/data/tables/time-series/demo/popest/intercensal-2010-2020-counties.html (2022)
19. United States Census Bureau: County intercensal tables: 2000-2010. https://www.census.gov/data/tables/time-series/demo/popest/intercensal-2000-2010-counties.html (2022)
20. Valenzuela, N., Báñuelos, M.: The role of california fires in predicting valley fever. In: Proceedings of the International Conference on Pattern Recognition. pp. 1–10 (2023)